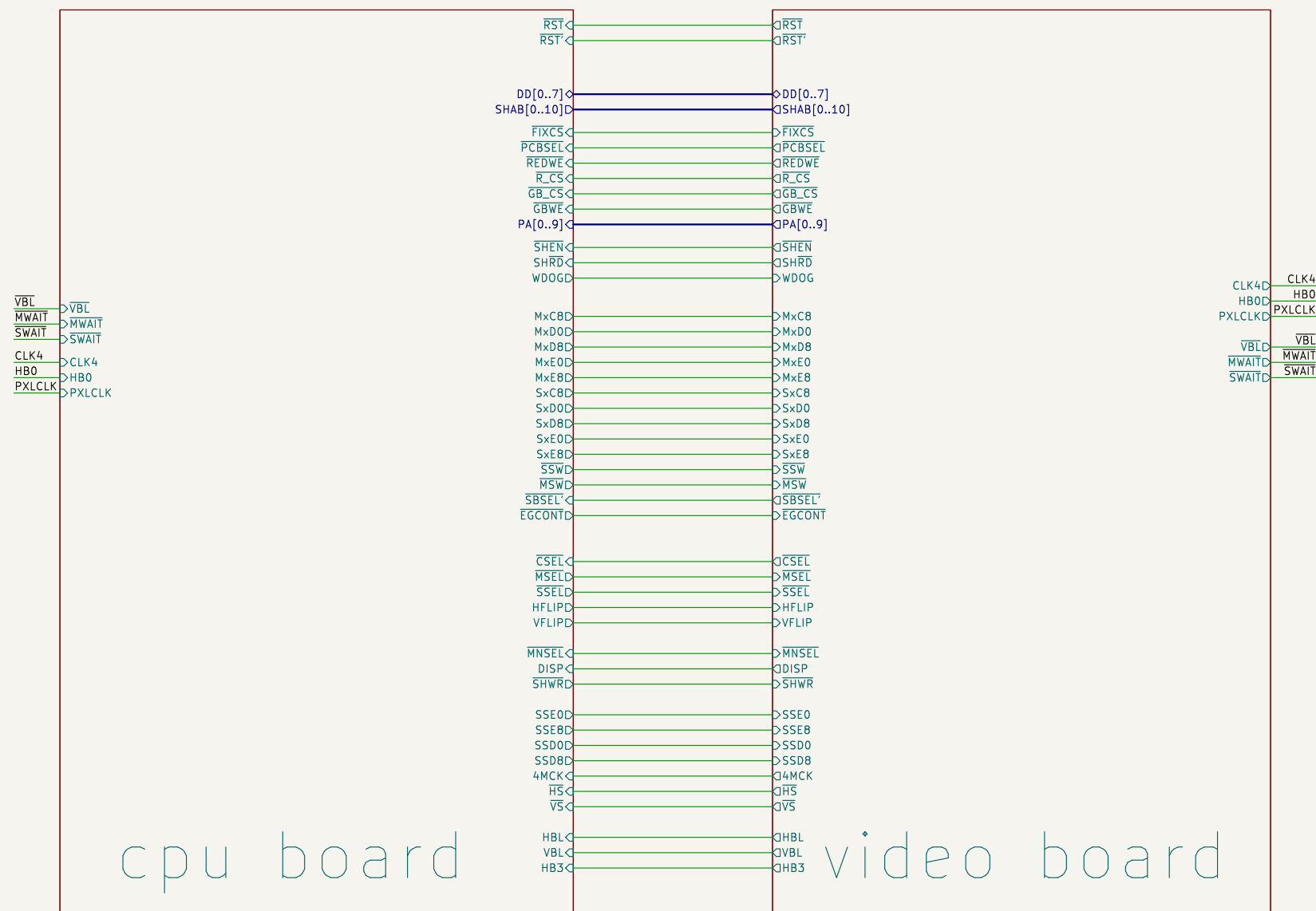




cpu board

video board

```
tehkanwch
bootleg board
```

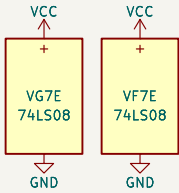
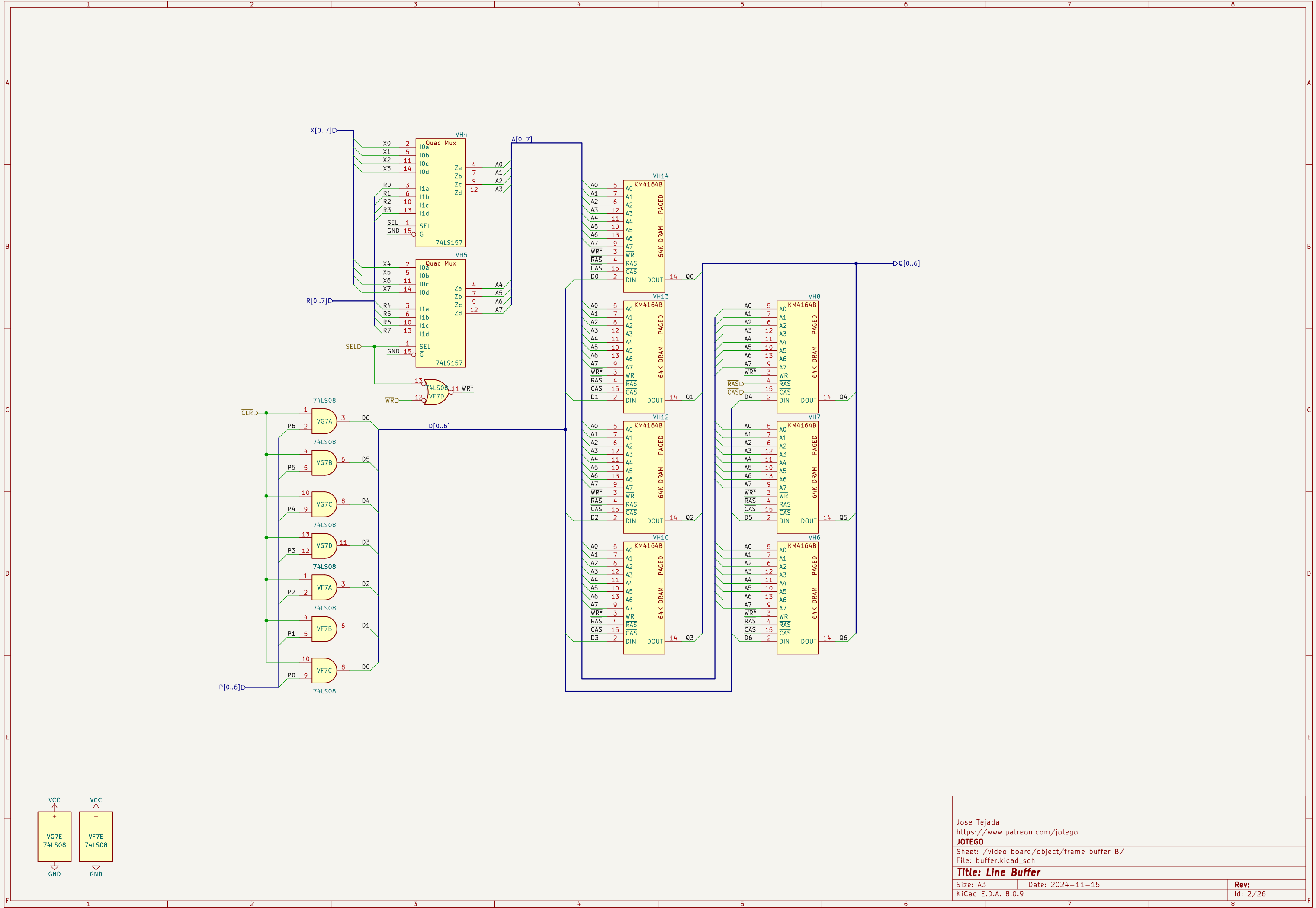
- original PCB has through-hole vias whereas the bootleg has filled vias
 - original uses three custom packages from Mitsubishi with logic inside.
- These were replaced by a small daughter board on the bootleg, serving the same functionality
- different edge connectors
 - debug edge connector on video board not present on bootleg
 - cabinet input logic partly missing and rewired on the bootleg to replace the trackball interface for a joystick one
 - different power amplifier for sound without heat sink on the bootleg
 - slightly different frequencies for crystal oscillators (cost reduction?)
 - device reference markers (silkscreen) on bootleg have reversed order on bootleg

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 Sheet: /
 File: tehkanwch.kicad_sch



Title: TEHKAN WORLD CUP (tehkanwch bootleg)

Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 1/26



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Sheet: /video board/object/frame buffer B/
File: buffer.kicad_sch

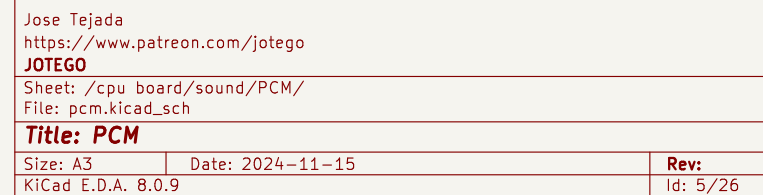
Title: Line Buffer

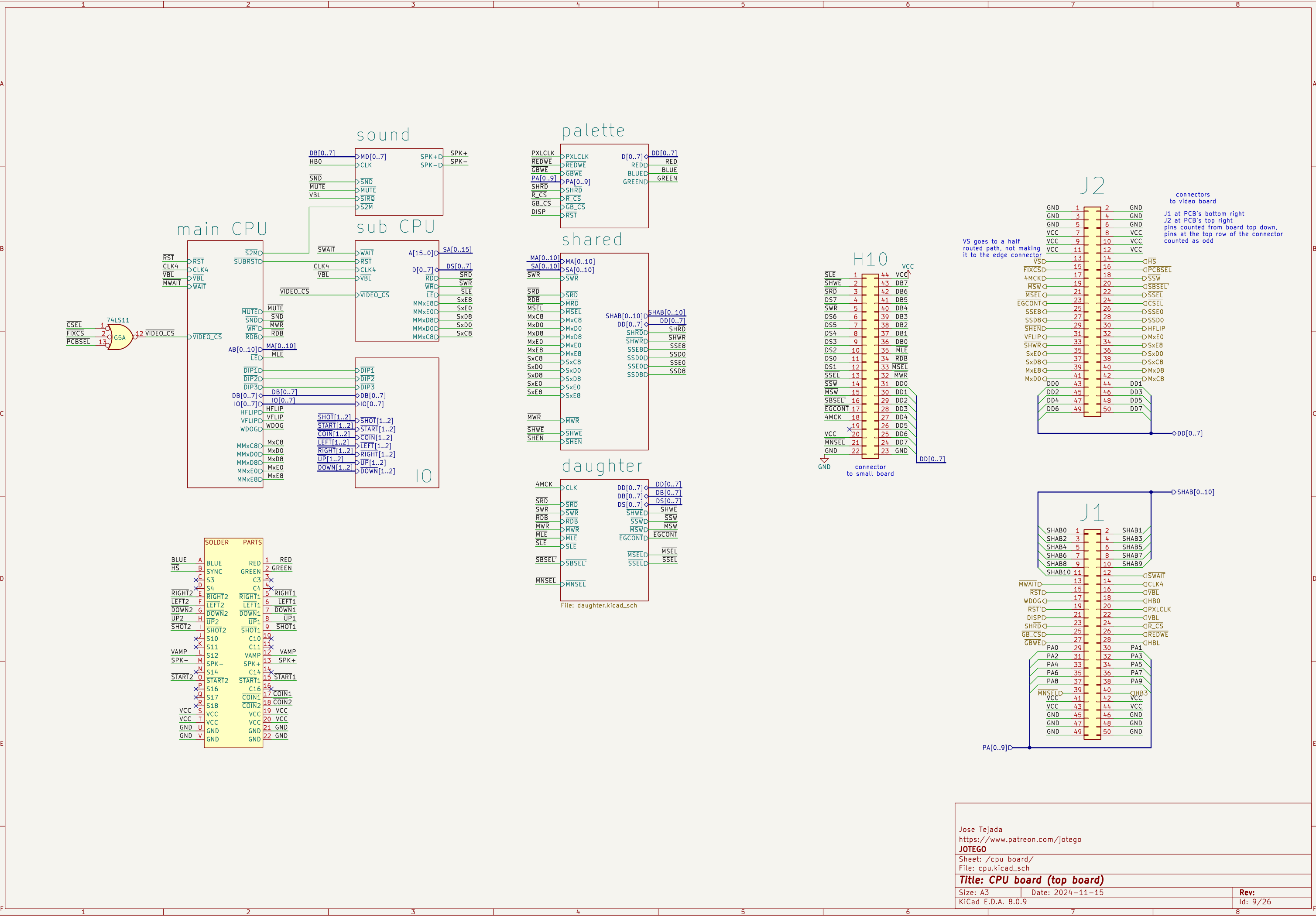
Size: A3 Date: 2024-11-15

KiCad E.D.A. 8.0.9

Rev:

Id: 2/26





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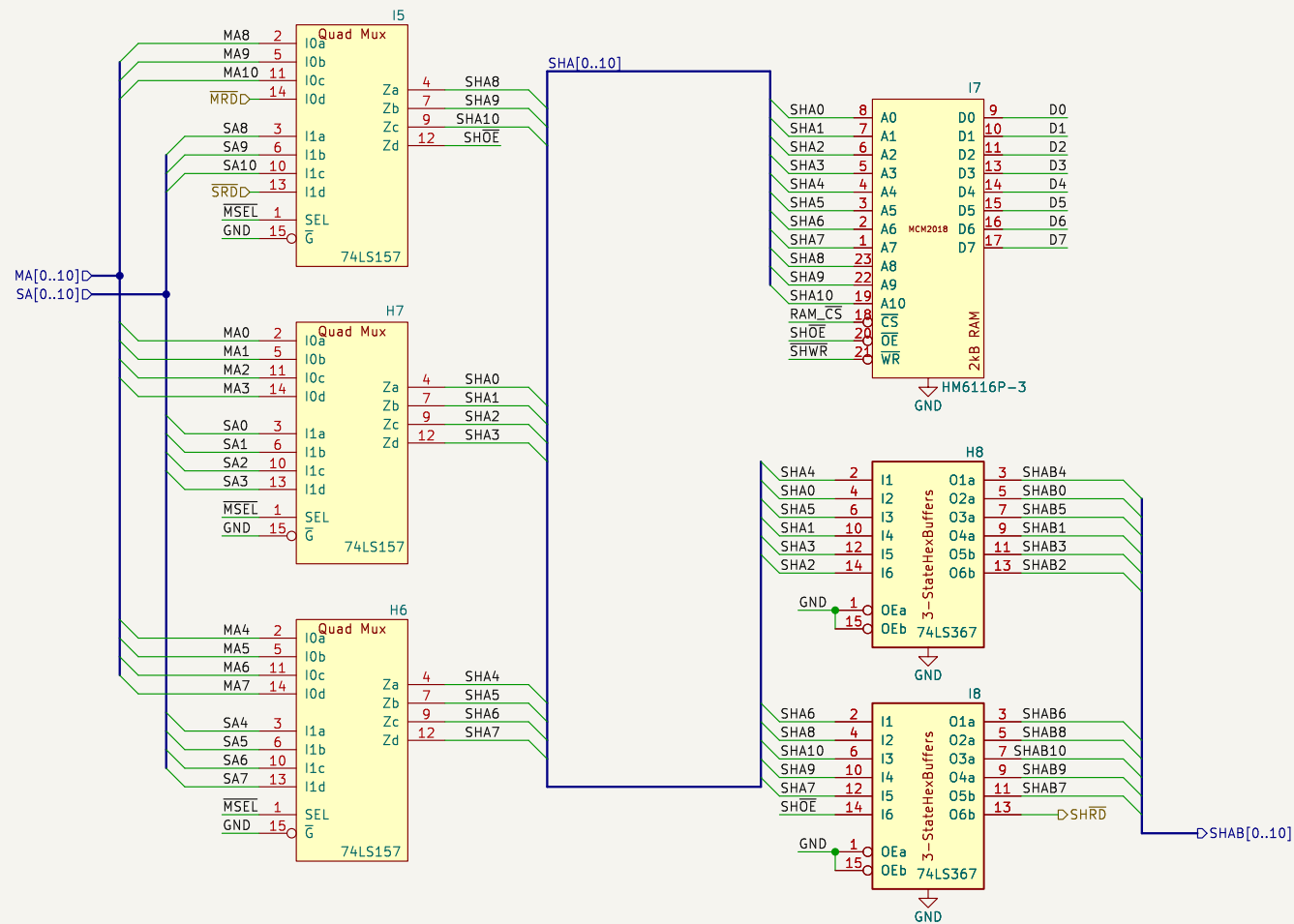
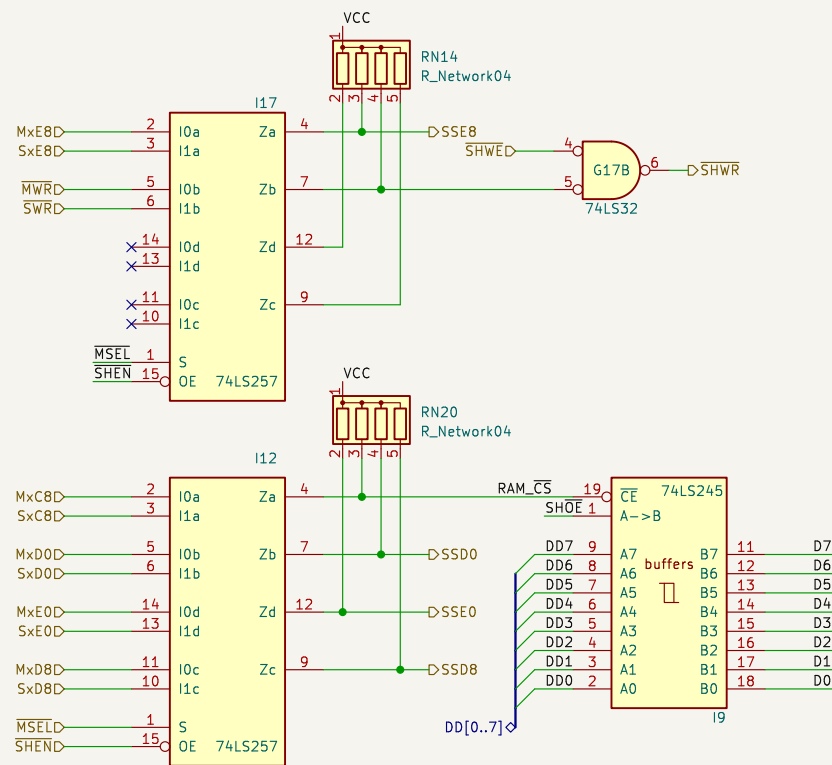
Sheet: /cpu board/
File: cpu.kicad_sch

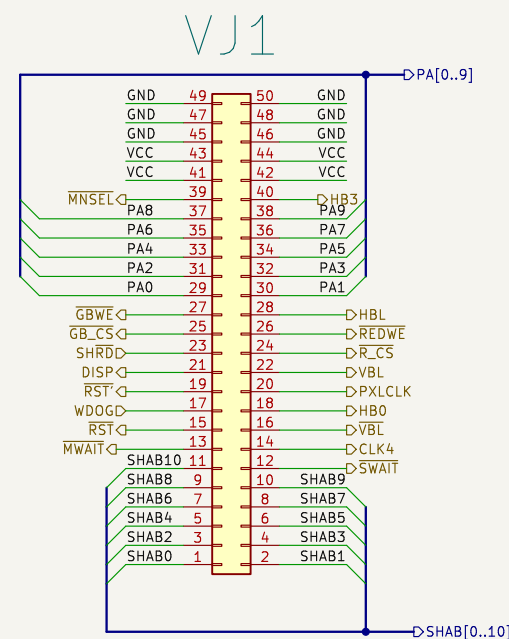
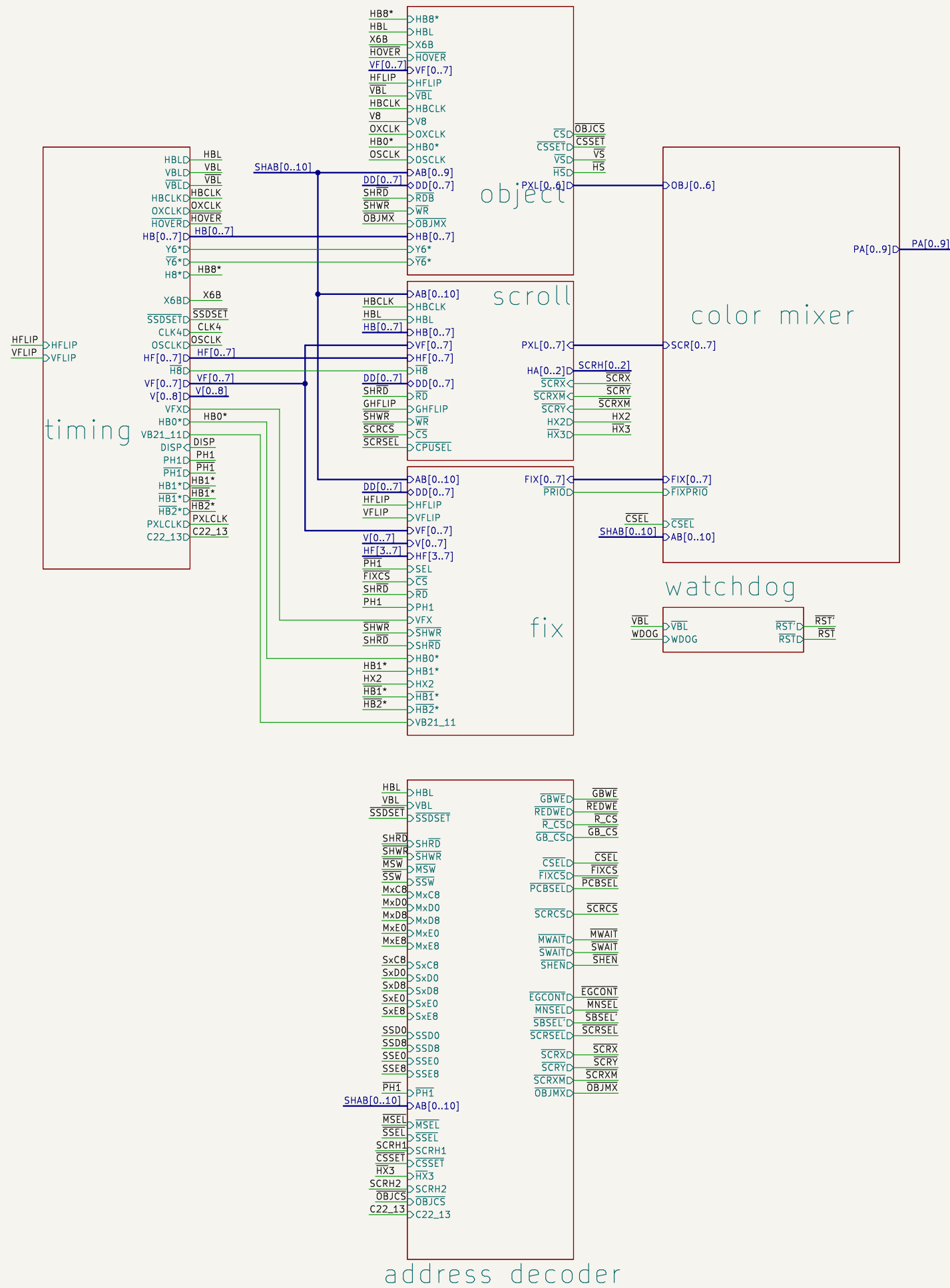
Title: CPU board (top board)

Size: A3 Date: 2024-11-15

KiCad E.D.A. 8.0.9

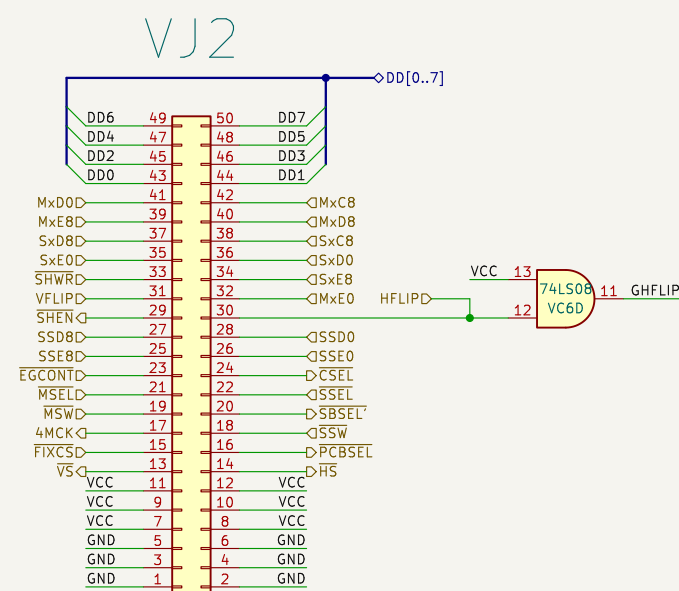
Rev:
Id: 9/26





connectors
to CPU board

VJ1 at PCB's top right
VJ2 at PCB's bottom right
pins counted from board bottom up,
pins at the top row of the connector
counted as even



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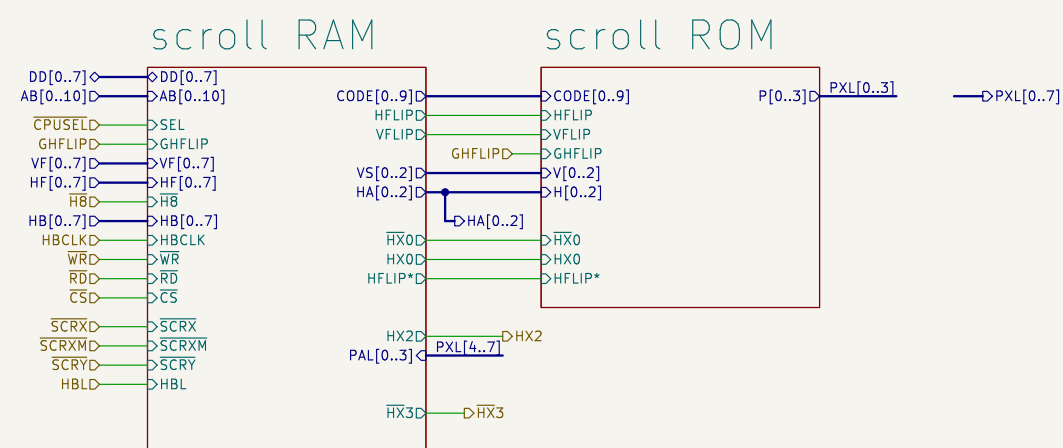
JOTEGO

Sheet: /video board/
File: video.kicad_sch

Title: Video board (bottom board)

Size: A3 Date: 2024-11-15
KiCad E.D.A. 8.0.9

Rev:
Id: 11/26

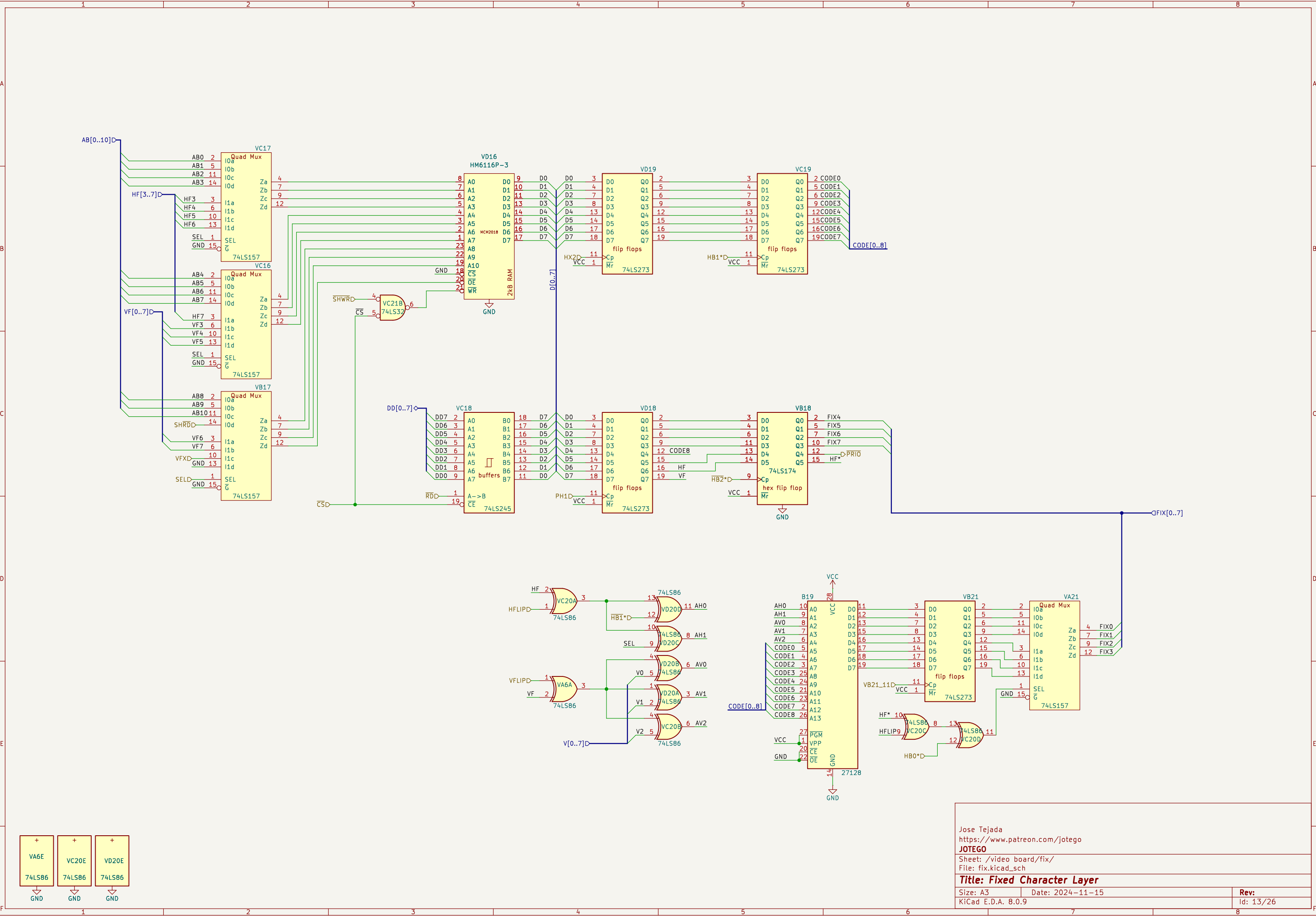


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 Sheet: /video board/scroll/
 File: scroll.kicad_sch

Title: Scroll Subsystem

Size: A3	Date: 2024-11-15
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Size: A3	Date: 2024-11-15	REV:
KiCad E.D.A. 8.0.9		Id: 12/26



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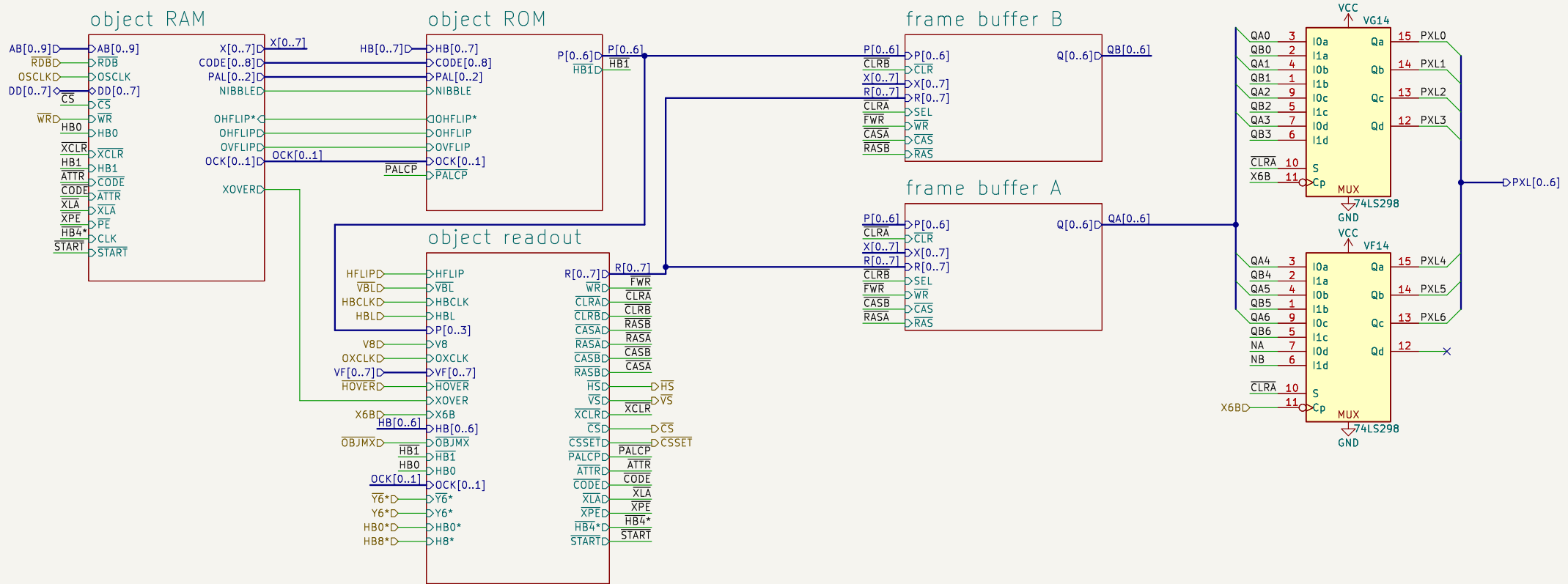
Sheet: /video board/fix/
File: fix.kicad_sch

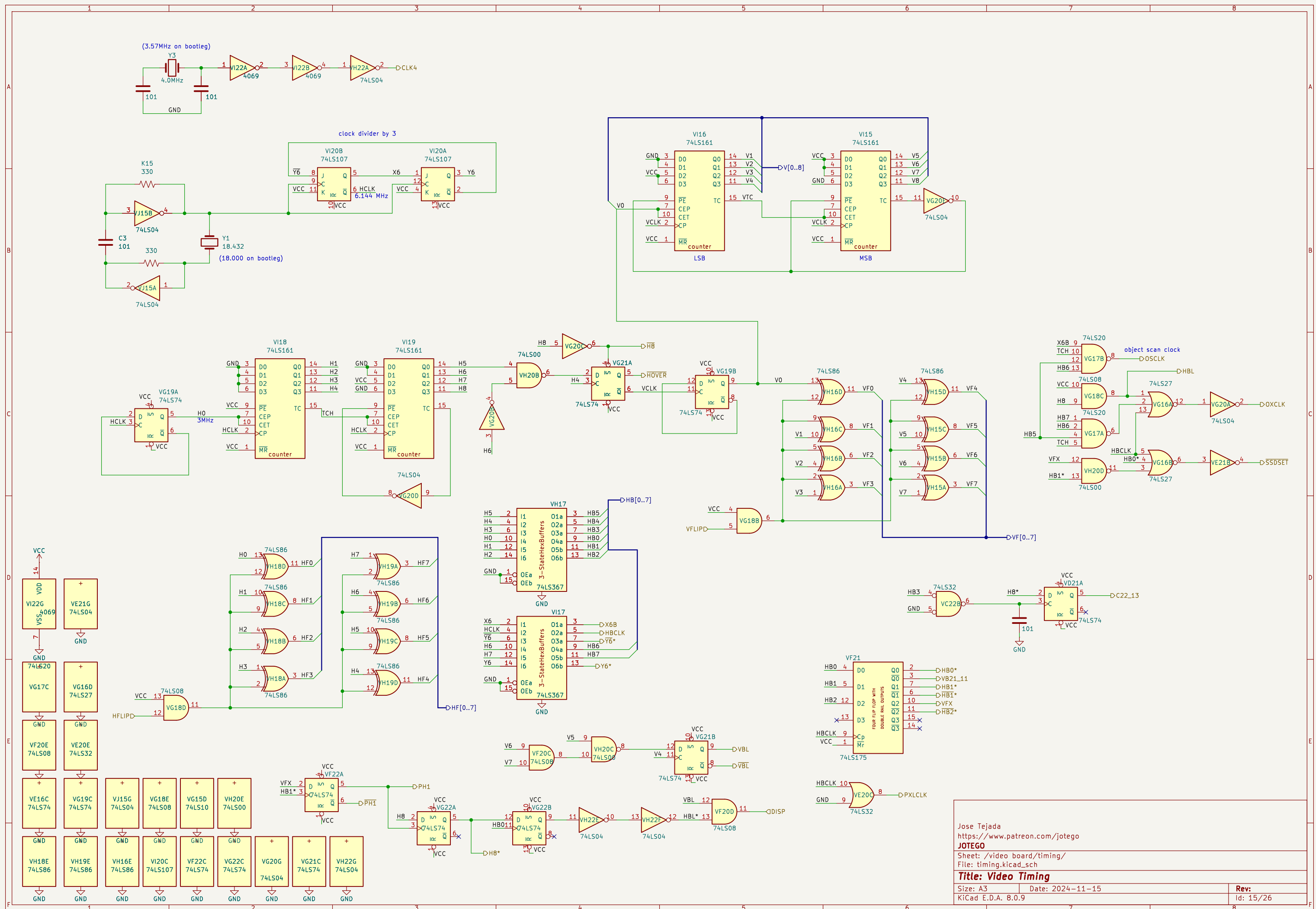
Title: Fixed Character Layer

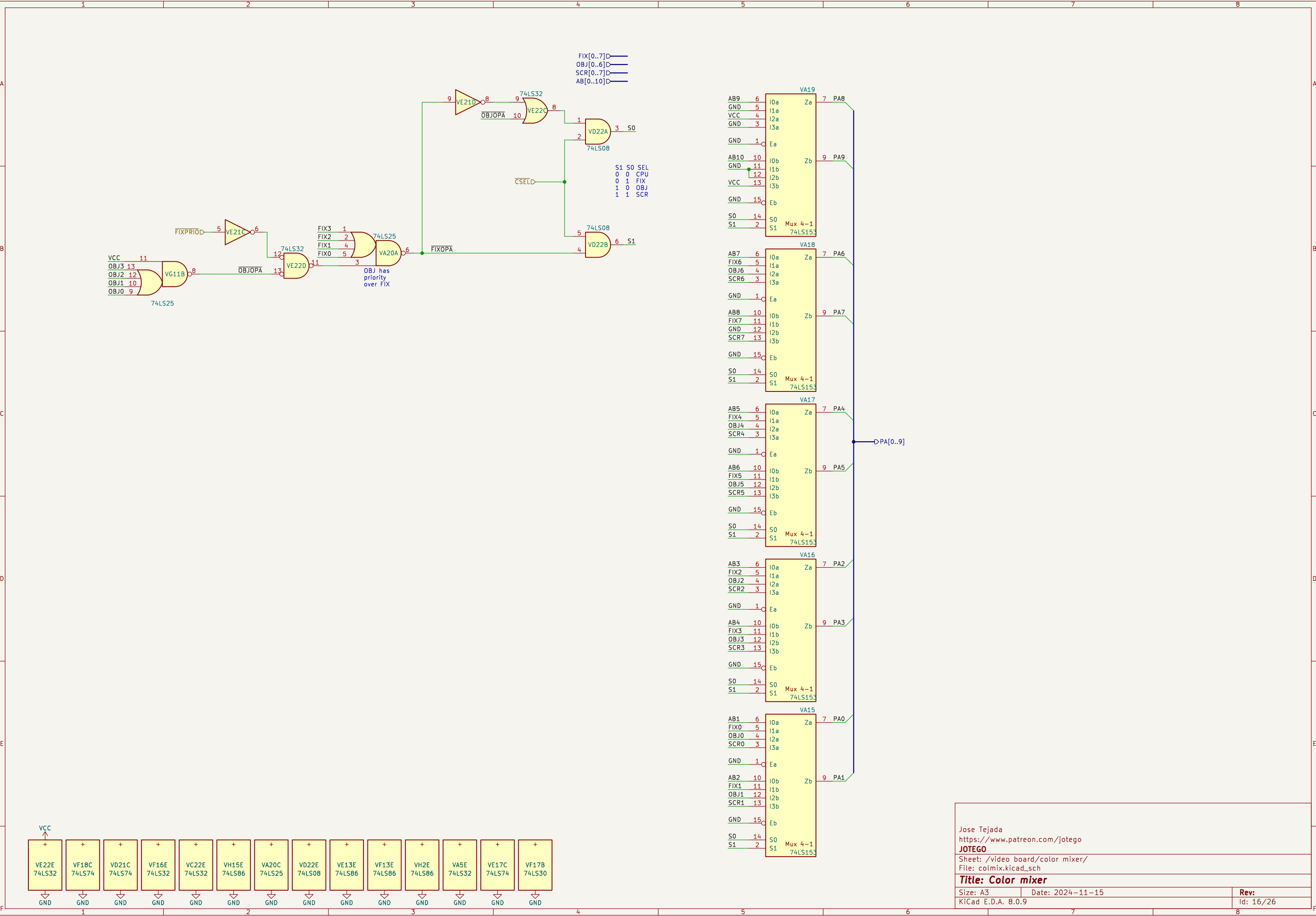
Size: A3 Date: 2024-11-15

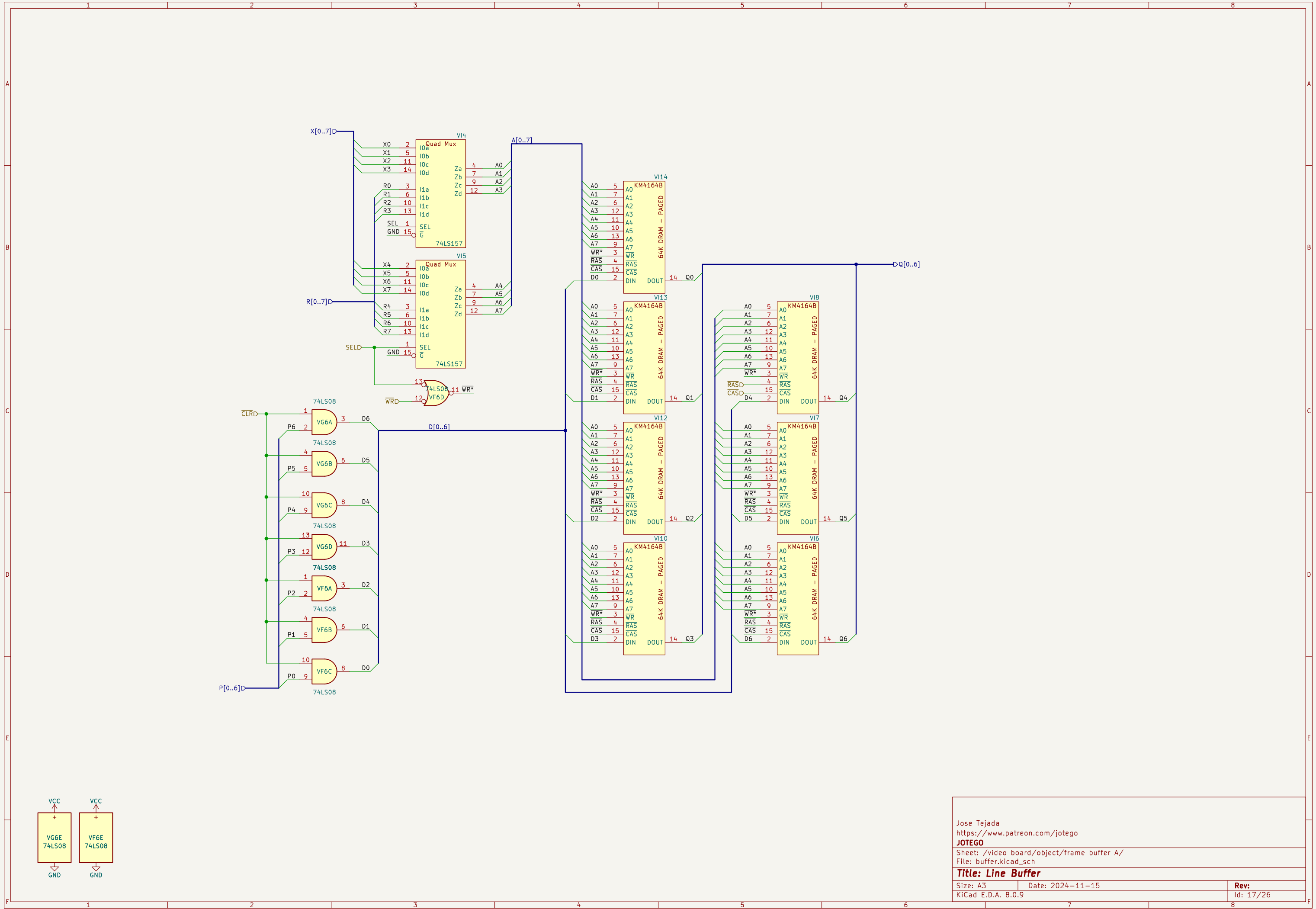
KiCad E.D.A. 8.0.9

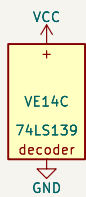
Rev:
Id: 13/26



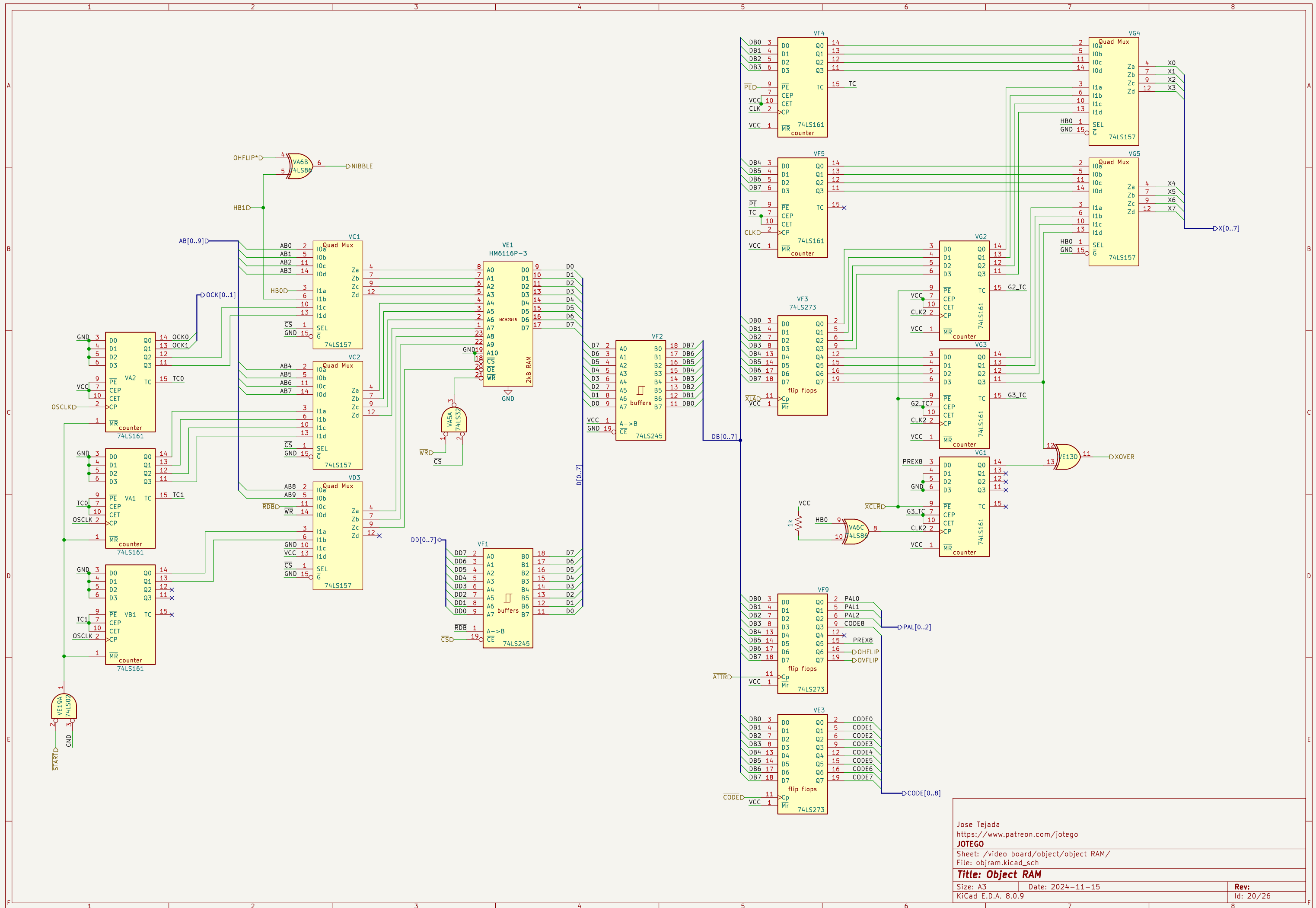








Rev:
Id: 19/26



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JOTEGO

Sheet: /video board/object/object RAM/
File: objram.kicad_sch

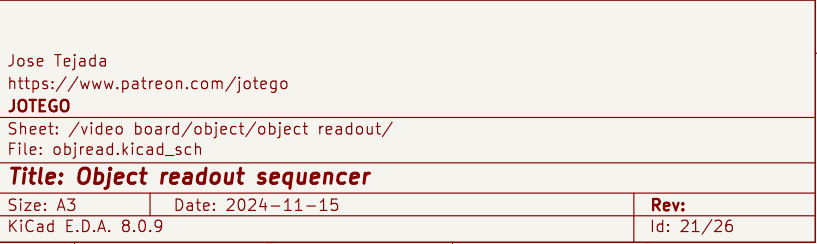
Title: Object RAM

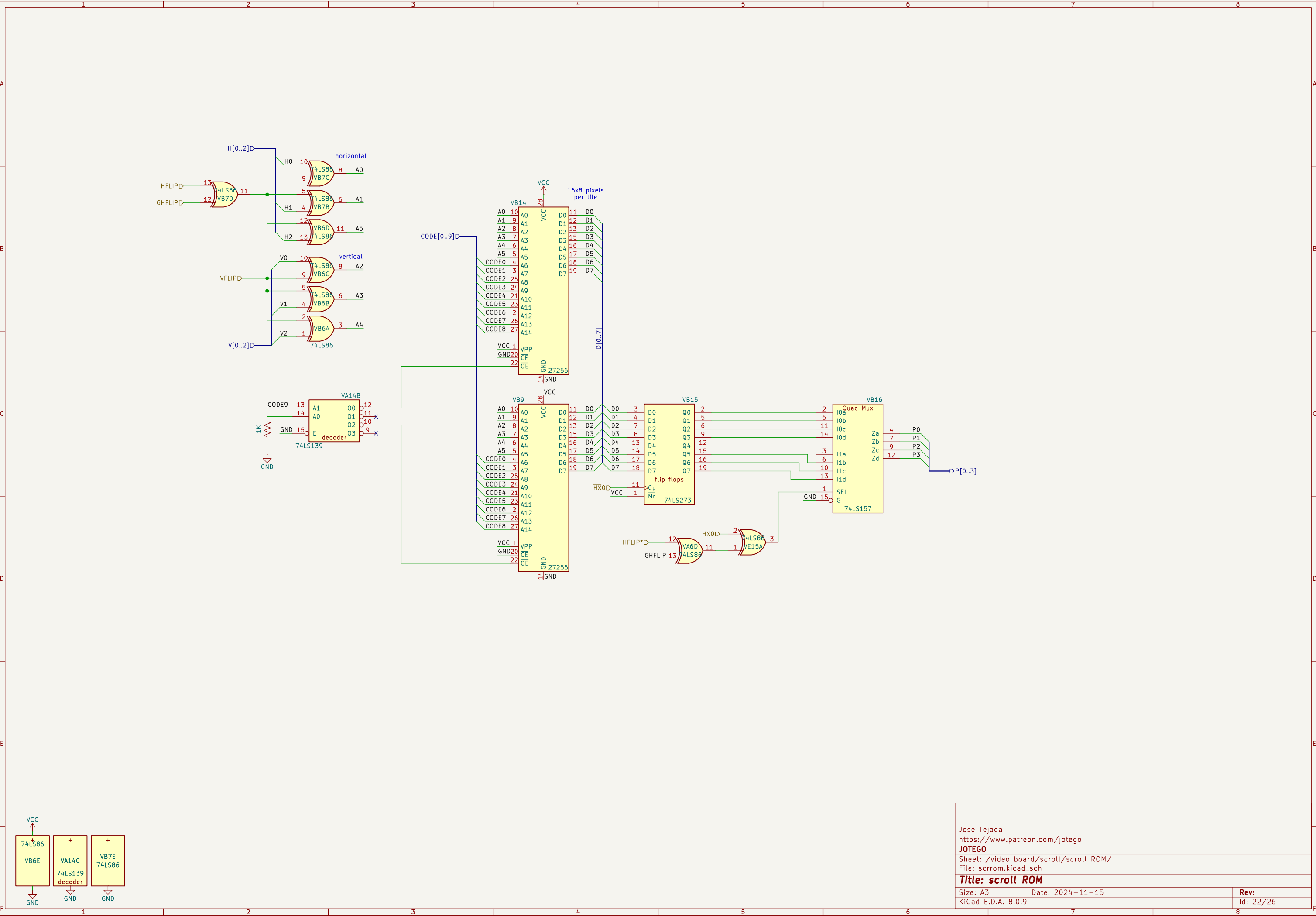
Size: A3

Date: 2024-11-15

Rev:

Id: 20/26





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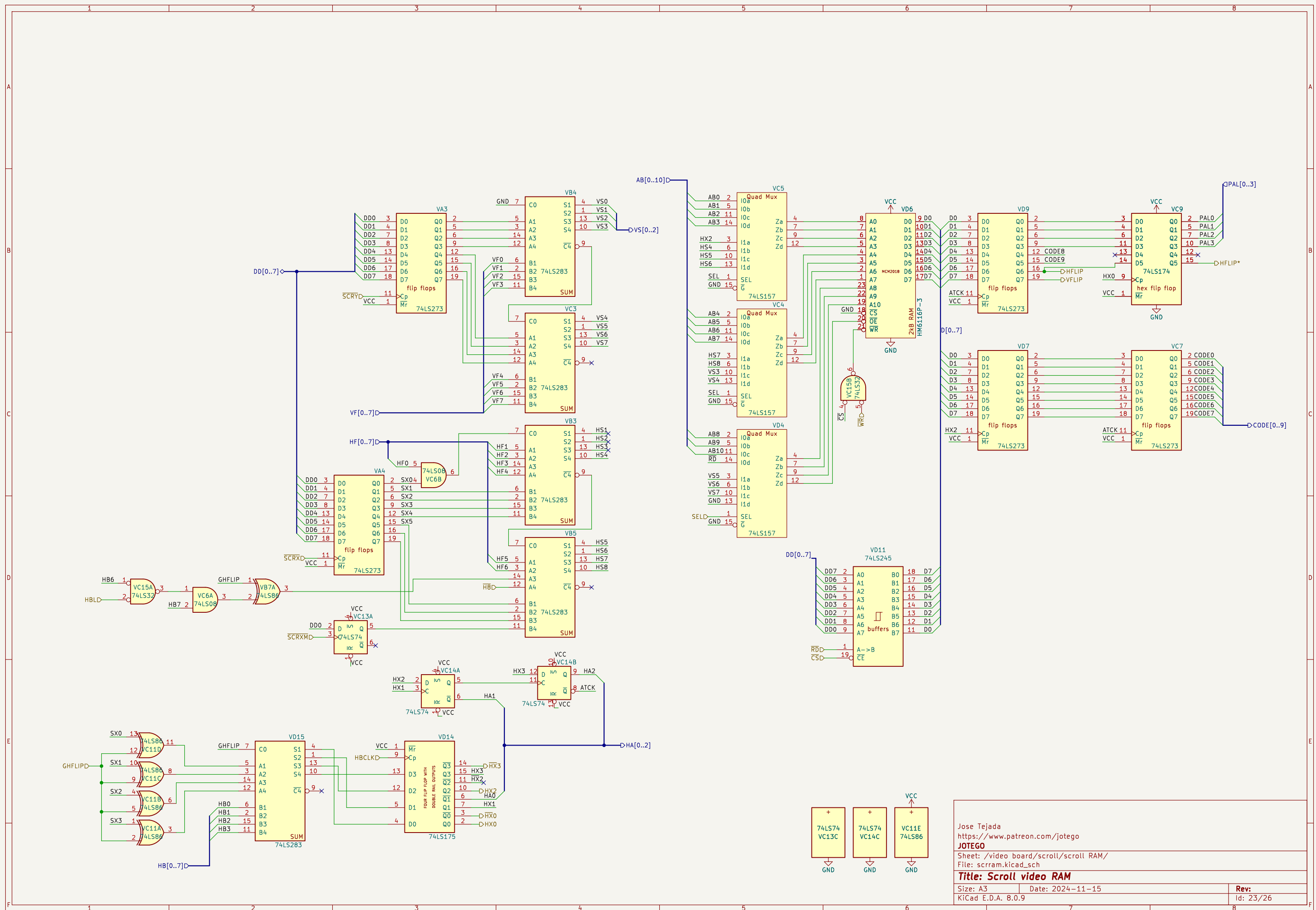
Sheet: /video board/scroll ROM/
File: scrrom.kicad_sch

Title: scroll ROM

Size: A3 Date: 2024-11-15

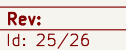
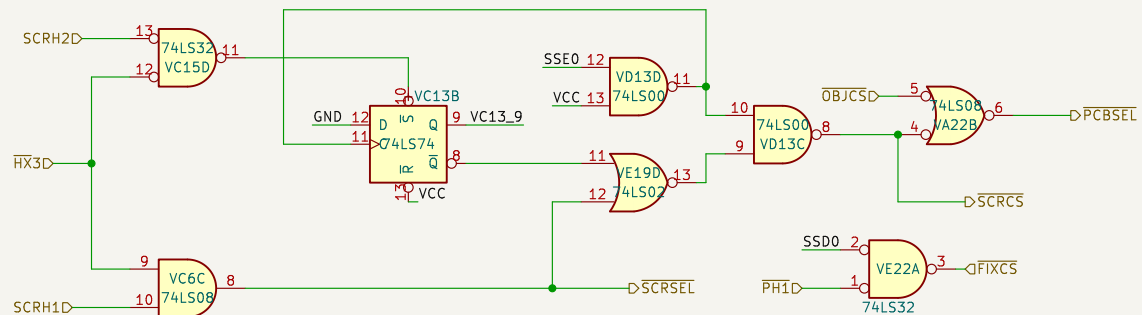
KiCad E.D.A. 8.0.9

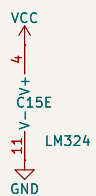
Rev:
Id: 22/26





Title: Watchdog Circuit		
Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 24/26

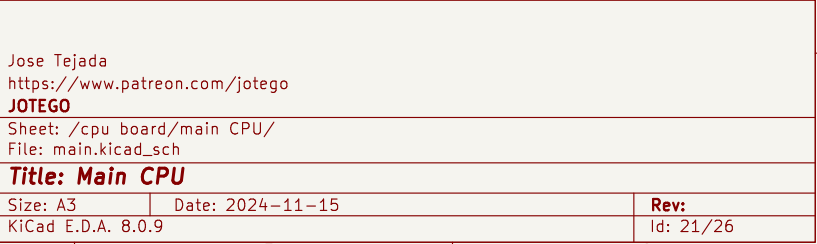


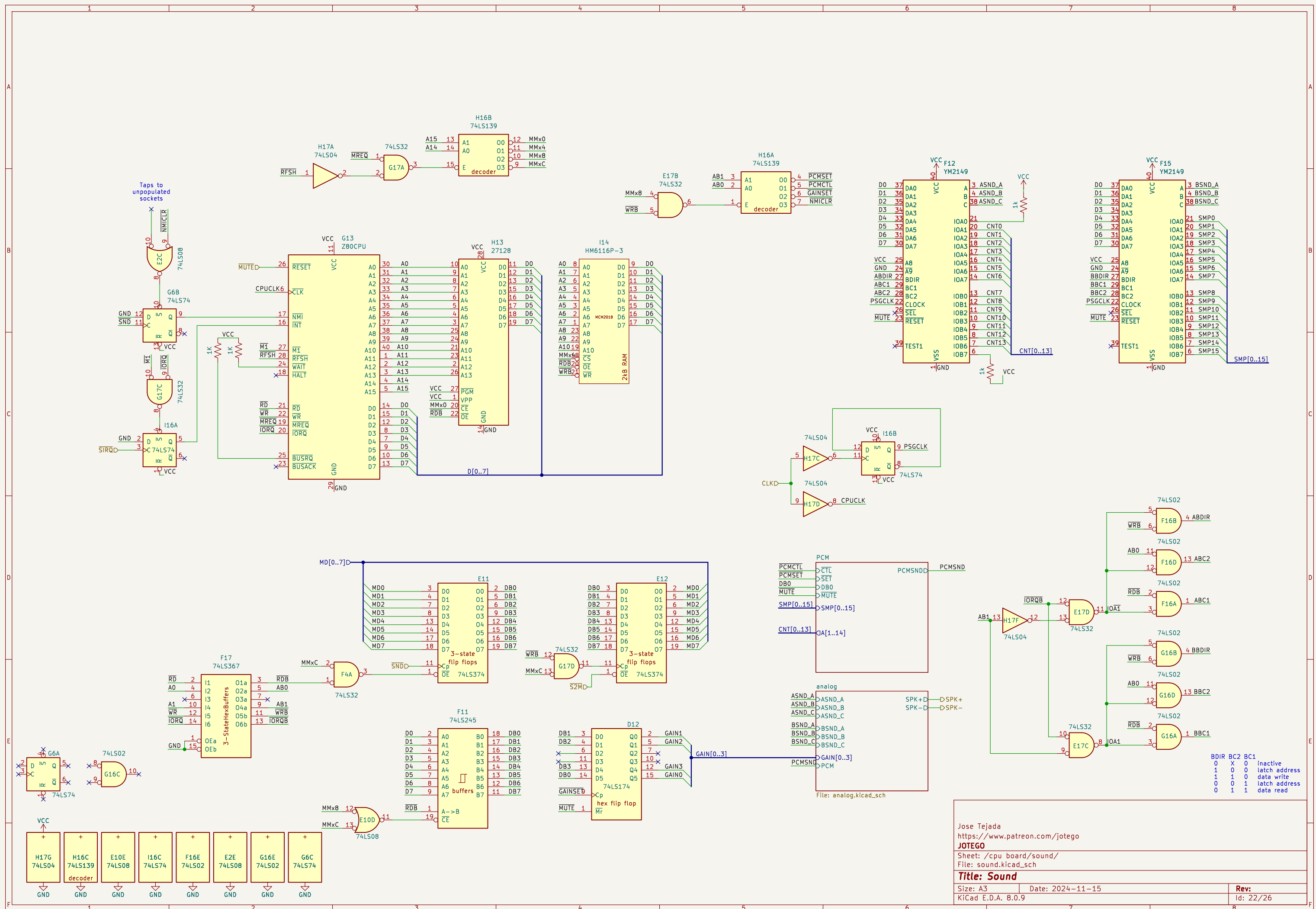


Device components follow bootleg references.
The bootleg counts letters from left to right and numbers from bottom to top
The original counts numbers left to right and letters bottom to top.

Title: Sound amplifiers

Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 26/26







Size: A3	Date: 2024-11-15
KiCad E.D.A. 8.0.9	

Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 23/26

